

Library Management System

(Project Proposal)

Project Code:

PGBH01-2003-RS

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1. Abstract

1.1. Overview of the Problem & Background:

Traditional educational institutions often rely on handwritten registers and paper-based tracking to manage library operations. This manual approach is highly prone to calculation errors, missing entries, data duplication, and physical wear, making it difficult for librarians to effectively track book stock, due dates, and outstanding collections.

1.2. Objective & Proposed Work Steps:

The primary objective of this project is to design and deploy a digital, computer-based Library Management System (LMS) to completely automate everyday library workflows. To achieve this, the project team will:

- Gather and document system specifications.(SRS)
- Design a normalized MySQL relational database schema.(make database)
- Develop a web-based user interface with explicit role-based access.
- Implement automated core modules for borrowing, returns, entering book, fine computations, and reservations of books.
- Conduct system testing and finalize the administrative documentation.(Test Plan)

1.3 Academic/Industrial Benefits (Significance): This system will replace inefficient paper-based workflows with a highly organized digital repository. It will minimize the time required for data entries, eradicate human errors in fine tracking, and enable quick asset searches for both students and staff. For the university campus, it establishes an auditable, high-performance local infrastructure to maintain administrative transparency and preserve intellectual assets safely.

2. Background and Justification

In reference to the institutional problem under study, educational centers require a structured framework to regulate resource distributions. Historically, libraries managed these allocations through manual ledgers. While some basic electronic workarounds exist, they often run on disjointed desktop platforms that lack multi-user concurrency or unified constraint checks.

This project enhances existing operational methodologies by replacing outdated manual records with a centralized Database Management System (DBMS) running over a Local Area Network (LAN). The justification for this system lies in its ability to enforce strict borrowing rules dynamically (e.g., specific limits for students versus teachers), compute fine metrics automatically without manual manipulation, and prevent data collisions when multiple operators log transaction entries at the same time. This transition directly minimizes administrative friction and ensures high data reliability.

3. Project Methodology

To successfully build the Library Management System, the project team will follow an agile and structured software engineering process divided into consecutive operational phases:

1. **Requirements Consolidation:** Analyzing user personas (Librarians, Students, and Teachers) to solidify system parameters and finalize the baseline layout.
2. **Database & Architecture Design:** Developing a robust back_end relational structure using MySQL Server to securely host primary asset tables .
3. **Front_end & Interface Development:** Building responsive, localized user screens utilizing modern web browsers or application wrappers to ensure a clean, intuitive layout for daily data entries.
4. **Module Integration:** Writing core logic modules for explicit functional controls, including Book Issuing constraints, Return Processing histories, and Reservation priority queues.
5. **System Validation & Testing:** Running data accuracy, authentication boundary filtering, and local backup persistence tests to guarantee system stability and prevent critical data loss.
6. **Deployment & Documentation:** Configuring the software package over the local on-premise network topology and delivering administrative training manuals to the librarians.

4. Project Scope

What the system will do (In-Scope Functionality):

- **Asset Tracking:** Provide comprehensive tools to add, update, search, and remove unique book records containing titles, authors, ISBNs, and tracking parameters.
- **Profile Management:** Maintain separate profile registries and access permission layers for students, teachers, and system administrators.
- **Rule Enforcement:** Enforce precise borrowing logic, limiting active loans to a maximum of 5 books for 10 days for students, and 8 books for 20 days for teachers/admins.
- **Automated Computations:** Automatically track overdue book counts, update dynamic inventory statuses post-return, and compute fines at a rate of 5 rupees per day.
- **Queue Notifications:** Manage a reserved book queue system and issue advanced reminders two days before due dates via a local synchronization framework.
- **Administrative Audit:** Generate consolidated activity logs and reports covering active checkouts, overdue items, and outstanding/paid fine balances.

What the system will not do (Out-of-Scope Boundaries):

- **No Direct Payment Processing:** The system will track, calculate, and log fine ledgers but will not directly handle or process external electronic/financial payment gateway transactions.
- **No Inter-Library Networking:** The architecture will only operate within the local campus infrastructure ecosystem and will not facilitate book loans with external outside libraries.

5. High level Project Plan

The project will be executed in multiple phases, each with defined deliverables and timelines.

Project Phases

- Requirement Analysis & SRS Preparation – 1 Week
- Database Design (MySQL Schema) – 1 Week
- UI/Frontend Development – 2 Weeks
- Backend Development & Module Integration – 3 Weeks
- Testing & Quality Assurance – 1–2 Weeks
- Deployment & Documentation – 1 Week

6. References

Resources includes:

- Software Engineering textbooks (e.g., Pressman, Roger S.)
- Database System Concepts – Abraham Silberschatz, Henry Korth
- Research journals on Library Management Systems and DBMS applications
- Online documentation for MySQL and web development frameworks
- Academic websites and technical blogs related to LMS design and implementation